BOAMP Grand Ouest Renewal-Linkage Project

Historical Session Documentation, Metrics, and Proposed Next Pipeline

Prepared for the BOAMP Data Science Internship Project

11 August 2026

Document Status

Superseded historical report. This document records the project state on 11 August 2026, before the current v2 standardisation, episode reconstruction, full-cohort linkage, Fellegi–Sunter comparison, expiry-aware sensitivity arm, and survival analysis were completed. Statements below using "current", "not yet", or "next" refer to that historical session and must not be read as the present repository state. The current methodology is documented in `reports/boamp_methodology_chapter.tex`; the expiry extension is documented in `reports/expiry_aware_successor_selection_v1.tex`.

The raw EDA in this historical branch counts literal `nomacheteur` strings. Consequently `Ville de Paris` and `VILLE DE PARIS` appear as different buyers in its chart. This is a property of the raw source and the exact-string EDA, not of the current v2 identity layer, which casefolds both to `ville de paris`. Longer organisational suffixes still require SIREN or other entity-resolution evidence.

Technical Summary

This document consolidates a historical working session on the BOAMP public procurement project. The project objective is to study renewal behavior of digital public procurement contracts in Grand Ouest using BOAMP data. BOAMP does not provide an explicit ground-truth field stating that one contract renews another, so the project requires a record-linkage stage before survival analysis.

The recommended methodological position is conservative:

> Use BOAMP to identify high-confidence, observable renewal links. Do not claim to recover every true renewal in the procurement market.

The current project status is:

Phase	Status	Main Artifact
Raw BOAMP acquisition	Done	`data/raw/boamp/`
Grand Ouest filtering	Done	`boamp_2015_2025_grand_ouest_raw.csv`
EDA	Done	`notebooks/boamp_full_raw_eda.ipynb`
Preprocessing / engineering	Done	Notebook 03
Manual reference benchmark	Done	Notebook 04
Candidate pair generation	Done for benchmark anchors	Notebook 05
Baseline linkage comparison	Done	Notebook 06
High-precision iteration	Done	Notebook 08
Parameter diagnostics	Done	Notebook 09
Full-corpus episode linkage	Not yet	Next implementation
Survival analysis	Not yet	After linkage is frozen

The current best linkage method is a transparent high-precision gated weighted score. On the locked test benchmark it improves precision and false-positive behavior compared with the first weighted baseline, but it is still not accurate enough to describe as a complete renewal detector.

Project Objective and Clean Flow

The real internship objective is not simply to build many linkage algorithms. The project goal is:

> Study renewal behavior and future procurement needs for digital public contracts in Grand Ouest using BOAMP data.

The clean end-to-end flow is:

1. Acquire raw BOAMP notices for 2015–2025.
2. Restrict the analysis scope to Grand Ouest.
3. Perform exploratory data analysis.
4. Engineer buyer, date, text, CPV, amount, duration, and quality features.
5. Prepare a manually validated benchmark because BOAMP has no explicit renewal label.
6. Generate candidate successor pairs.
7. Compare simple linkage methods.
8. Tune parameters on `PILOT_DEVELOPMENT`.
9. Evaluate once on `LOCKED_TEST`.
10. Freeze one conservative linkage rule.
11. Apply the frozen rule to the full Grand Ouest digital-contract population.
12. Build renewal episodes and survival-analysis data.
13. Run Kaplan-Meier, Cox, and sensitivity analyses.

External Methodological Context

The method follows standard record-linkage logic:

- The Fellegi-Sunter tradition treats linkage as cumulative evidence across several fields rather than a single exact identifier.[1]

[1] https://moj-analytical-services.github.io/splink/topic_guides/theory/fellegi_sunter.html

- Deterministic linkage is cheap and can achieve high precision, but usually has lower recall; probabilistic linkage allows the analyst to choose a threshold depending on the cost of false positives versus false negatives.[2]
- Blocking or indexing is necessary in large record linkage because comparing all possible pairs is computationally infeasible.[3]
- Active learning and clerical review are useful when ground truth is limited; a small number of carefully chosen manual labels can improve linkage performance.[4]
- Survival analysis is appropriate once a renewal event and time-to-event variable are defined. Right-censored observations should be kept, not dropped.[5]

BOAMP itself is the official source used for the raw notices.[6] CPV is an official hierarchical public procurement classification system, which justifies using CPV prefixes as evidence but not as the sole matching rule.[7]

Data Scope and Raw Acquisition

The main historical corpus covers:

$$2015\text{-}01\text{-}01 \leq \texttt{dateparution} < 2026\text{-}01\text{-}01$$

The Grand Ouest working scope is:

- Bretagne
- Normandie
- Pays de la Loire

The Grand Ouest raw CSV summary is:

Metric	Value
Full BOAMP rows scanned	1,620,712
Grand Ouest rows	226,314
Raw fields preserved	41
Unique `idweb`	226,314
Duplicate `idweb`	0
Missing `idweb`	0
Minimum `dateparution`	2015-03-02
Maximum `dateparution`	2025-12-31
Records outside 2015–2025 range	0

Rows by Grand Ouest region:

[2] https://moj-analytical-services.github.io/splink/topic_guides/theory/probabilistic_vs_deterministic.html

[3] https://openresearch-repository.anu.edu.au/items/29ab1c2e-6b1f-4d61-8691-185fed831565

[4] https://pmc.ncbi.nlm.nih.gov/articles/PMC10072450/

[5] https://pmc.ncbi.nlm.nih.gov/articles/PMC10357905/

[6] https://boamp-datadila.opendatasoft.com/explore/dataset/boamp/api/

[7] https://single-market-economy.ec.europa.eu/single-market/public-procurement/digital-procurement/common-procurement-vocabulary_en

Region	Notices
Bretagne	64,208
Normandie	86,180
Pays de la Loire	75,926

Rows by year:

Year	Notices	Year	Notices
2015	18,462	2021	20,827
2016	20,349	2022	21,349
2017	20,416	2023	22,012
2018	20,865	2024	20,507
2019	21,601	2025	20,754
2020	19,172	Total	226,314

Preprocessing and Engineered Dataset

The preprocessing notebook creates:

```
data/processed/boamp_grand_ouest/notices_engineered.parquet
```

It engineers identity, date, geography, buyer, text, CPV/theme, amount, duration, and quality features. Missing values are not blindly imputed. They are preserved and represented with explicit missingness flags.

Metric	Value
Engineered rows	226,314
Engineered fields	74
Unique `idweb`	226,314
Missing `idweb`	0
Duplicate `idweb`	0
Invalid `dateparution`	0
Outside historical range	0
Missing buyer key	10
Missing CPV	39,993
Digital candidate rows	24,869
Usable for linkage rows	226,299

Digital candidate rows by segment:

Segment	Rows
Telecom / network	8,655
IT services	7,337
Software	3,199
Cybersecurity	2,841
Hardware / infrastructure	1,479
Cloud / hosting	867
Data / AI	491

Manual Reference Benchmark

Because BOAMP does not include a renewal label, the reference benchmark acts like a train/test split in normal machine learning:

- `PILOT_DEVELOPMENT`: used to choose thresholds, gates, and weight presets.
- `LOCKED_TEST`: held out and used only once for final evaluation.

The current benchmark files are:

- `data/reference/BOAMP_Internship_Reference_120.csv`
- `data/reference/BOAMP_Internship_Evaluation_Subset.csv`
- `data/reference/BOAMP_Internship_Confirmed_Successor_Links.csv`

Benchmark summary:

Metric	Value
Reference anchors	120
Evaluation subset anchors	94
Confirmed successor link rows	28
`PILOT_DEVELOPMENT` anchors	16
`LOCKED_TEST` anchors	78
Anchor notice IDs missing in BOAMP	0
Successor notice IDs missing in BOAMP	0

Evaluation outcome distribution:

Outcome	Count
`NO_OBSERVED_SUCCESSOR_IN_SCOPE`	70
`OBSERVED_SUCCESSOR`	23
`MULTIPLE_SUCCESSORS`	1

Candidate Pair Generation

Candidate generation is currently benchmark-focused. It creates possible future successors for reference anchors:

```
data/processed/boamp_grand_ouest/renewal_candidate_pairs.parquet
```

Candidate rules:

- candidate date must be after anchor date;
- time gap between roughly 3 months and 8 years;
- same buyer key or similar normalized buyer name;
- same CPV/theme or strong text similarity;
- Grand Ouest only.

Candidate generation metrics:

Metric	Value
Candidate pairs	8,095
Reference anchors covered	120 / 120
Eligible evaluation anchors covered	94 / 94

Linkage Methods Tested

The following methods were implemented and compared:

Method	Explanation
M0: no-link baseline	Always predicts no successor. This is a sanity baseline.
M1: exact-rule baseline	Accepts strict deterministic matches based on buyer and timing.
M2: nearest valid successor	Chooses the nearest future valid candidate within the allowed time window.
M3: weighted rule score	First main baseline. Uses weighted buyer, text, CPV, time, and geography evidence, then applies a threshold.
M4: word TF-IDF similarity	Ranks candidates mainly by word-level TF-IDF similarity.
M5: char n-gram TF-IDF similarity	Ranks candidates mainly by character n-gram TF-IDF similarity.
M6: high-precision gated score	Current best direction. Uses a weighted score plus explicit acceptance gates to reduce false positives.

Linkage Metrics

The evaluation uses the following metrics:

Metric	Meaning
Coverage rate / linking rate	Share of anchors for which the model predicts at least one successor.
Precision@1	Among anchors with a predicted top successor, share where the top prediction is correct. This is the most important automatic-link quality metric.
Recall@1	Among anchors that truly have a successor in the benchmark, share where the true successor is found as top-1.
Recall@5	Among anchors that truly have a successor, share where the true successor appears anywhere in the top 5 candidates. Useful for manual review.

Metric	Meaning
Mean reciprocal rank	Rewards ranking the true successor near the top of the candidate list.
False-positive rate on no-successor anchors	Among anchors manually labelled as no observed successor, share where the model wrongly predicts a successor. This matters strongly for survival analysis.
No-link accuracy	Among no-successor anchors, share where the model correctly predicts no link.
Predicted count	Number of anchors where the model emits an automatic link.

Baseline Versus High-Precision Results

The first useful baseline was `M3_weighted_rule_score`. It was too permissive: it linked more anchors, but many predicted renewals were false positives.

Metric on `LOCKED_TEST`	M3 baseline	M6 high precision
Anchors	78	78
Positive successor anchors	19	19
No-successor anchors	59	59
Coverage rate	0.564	0.410
Precision@1	0.250	0.438
Recall@1	0.579	0.737
Recall@5	0.684	0.842
Mean reciprocal rank	0.152	0.190
False-positive rate	0.441	0.237
No-link accuracy	0.559	0.763
Predicted count	44	32

Interpretation:

- The high-precision method improves **precision@1** from 0.250 to 0.438.
- It reduces **false-positive rate** from 0.441 to 0.237.
- It lowers **coverage** from 0.564 to 0.410.
- This is the expected tradeoff: fewer links, but more reliable links.

Current Linking Rule

The current best rule is:

```
M6_high_precision_gated_score
```

It computes:

$$\text{score} = 100 \times (0.50 \times \text{buyer} + 0.25 \times \text{text} + 0.15 \times \text{CPV} + 0.05 \times \text{time} + 0.05 \times \text{geo})$$

The selected setting is:

Parameter	Value
Weight preset	`buyer_precision_v1`
Buyer weight	0.50
Text weight	0.25
CPV/theme weight	0.15
Time plausibility weight	0.05
Geography weight	0.05
Automatic threshold	65
Minimum text similarity	0.25
Text override for missing CPV	0.55
Minimum fuzzy-buyer text similarity	0.45
Minimum gap	180 days
Maximum gap	2,920 days

Acceptance requires:

- strong buyer evidence: SIREN, normalized buyer name, or high fuzzy similarity;
- plausible future time gap;
- text similarity at least 0.25;
- CPV/theme continuity, unless text similarity is very high;
- score at least 65.

How the Weights Were Determined

The current weights were not learned automatically by a statistical model. They were determined by:

1. **Domain-informed design.** Buyer continuity should matter most for procurement renewal. Text similarity should matter strongly because the object of the contract identifies the need. CPV/theme should matter, but CPV can be noisy. Time matters, but duration is often missing. Geography matters least because the data is already restricted to Grand Ouest.
2. **Pilot benchmark selection.** Several weight presets and thresholds were tested only on `PILOT_DEVELOPMENT`. The selected setting was then evaluated on `LOCKED_TEST`.

Therefore the correct description is:

> The linkage score is a transparent rule-based scoring model. Weights were specified from procurement-domain reasoning and selected through pilot-set tuning, then evaluated on a locked test subset.

With more manually labelled pairs, future versions could estimate weights statistically using logistic regression, a Fellegi-Sunter model, gradient boosting, or active learning. With the current benchmark size, a transparent rule is more defensible than a complex model.

Parameter Diagnostics

A dedicated parameter-selection notebook was created:

```
notebooks/09_linkage_parameter_selection_diagnostics.ipynb
```

It generated:

- `data/processed/boamp_grand_ouest/parameter_selection_grid.csv`
- `data/processed/boamp_grand_ouest/parameter_selection_recommended_config.json`
- six figures under `data/processed/boamp_grand_ouest/figures/parameter_selection/`

The current setting and diagnostic recommendation had the same pilot metrics:

Pilot metric	Current setting	Grid recommendation
Coverage rate	0.3125	0.3125
Precision@1	0.800	0.800
Recall@1	0.800	0.800
Recall@5	1.000	1.000
False-positive rate	0.000	0.000
No-link accuracy	1.000	1.000
Predicted count	5	5

The grid recommendation relaxed minimum text similarity from 0.25 to 0.20 on the pilot split, but because the pilot set contains only 16 anchors, the safer practical choice is to keep the current 0.25 threshold unless manual inspection supports relaxing it.

Why Precision Is Still a Concern

The locked-test high-precision **precision@1** is 0.4375. This means that among anchors where the model predicts a top successor, about 44% of top predictions are correct in the locked benchmark.

That is not good enough to blindly treat all predicted links as true renewals. However, it is still useful if the project is framed correctly:

> The method identifies a conservative subset of high-confidence observable renewals.
> It is not an exhaustive detector of all true renewals.

For survival analysis, false positives are dangerous because a wrong renewal link creates a false event time. Therefore, the project should prioritize precision over coverage.

Current Limitations

- The benchmark is small: 94 eligible anchors, with only 16 in the pilot split.
- Current candidate generation is benchmark-anchor focused, not yet full-corpus episode linkage.
- BOAMP notices are not the same as contract episodes; initial notices, awards, modifications, lots, and linked notices may fragment one procurement need.
- Buyer identifiers are incomplete; most buyer keys are normalized names rather than SIREN.
- CPV can be noisy and is not sufficient alone for renewal detection.
- Duration and amount fields are often missing, limiting time-plausibility evidence.
- Survival analysis is not ready until the final renewal-link table is created.

Recommended Optimal Next Pipeline

The most efficient next step is not to add many more algorithms. The next implementation should move toward the real research objective:

1. Build procurement episodes from engineered notices.
2. Restrict to Grand Ouest digital episodes.
3. Generate candidate successor episodes for the full digital population.
4. Apply the frozen high-precision linkage rule.
5. Create `final_renewal_links.parquet`.
6. Create `survival_episodes.parquet`.
7. Run survival analysis on high-confidence identifiable renewals.
8. Add sensitivity analysis using stricter and looser linkage thresholds.

The most important methodological improvement is:

> Do not link raw notices directly if possible. First construct procurement episodes, then link episodes.

Future Survival Metrics

After linkage is frozen, survival analysis should use:

- renewal observed indicator;
- time to renewal in months;
- right-censoring indicator;
- renewal probability within 12, 24, and 36 months;
- Kaplan-Meier survival curves by technology segment;
- log-rank tests across segments;
- Cox proportional hazards model coefficients and hazard ratios;
- sensitivity analysis across linkage thresholds.

Important Local Artifacts

Most processed artifacts are under:

```
data/processed/boamp_grand_ouest/
```

Purpose	File name
Engineered notices	`notices_engineered.parquet`
Reference anchors	`reference_anchor_episodes.parquet`
Reference links	`reference_successor_links.parquet`
Candidate pairs	`renewal_candidate_pairs.parquet`
Baseline predictions	`linkage_baseline_predictions.parquet`
Baseline evaluation	`linkage_evaluation_results.csv`
High-precision predictions	`high_precision_linkage_predictions.parquet`
High-precision evaluation	`high_precision_linkage_evaluation.csv`
High-precision config	`high_precision_linkage_config.json`
Parameter grid	`parameter_selection_grid.csv`
Parameter recommendation	`parameter_selection_recommended_config.json`
Manual review candidates	`high_precision_review_candidates.csv`

Figure Folders

The session produced three main diagnostic figure folders:

- `data/processed/boamp_grand_ouest/figures/linkage_baselines/`
- `data/processed/boamp_grand_ouest/figures/high_precision_linkage/`
- `data/processed/boamp_grand_ouest/figures/parameter_selection/`

The parameter-selection figures are:

- `01_pilot_threshold_metric_curves.png`
- `02_pilot_precision_coverage_frontier.png`
- `03_pilot_precision_heatmap_threshold_text.png`
- `04_pilot_false_positive_heatmap_threshold_text.png`
- `05_pilot_recall5_heatmap_threshold_text.png`
- `06_pilot_current_vs_recommended_metrics.png`

Recommended Report Framing

For the internship report, use this framing:

> Because BOAMP does not provide explicit renewal labels, the study uses a conservative record-linkage approach. Candidate successor contracts are generated through temporal, buyer, CPV/theme, and text-based blocking. A transparent high-precision scoring rule is tuned on a pilot manual benchmark and evaluated on a locked test benchmark. The resulting links are interpreted as identifiable high-confidence renewals rather than exhaustive ground truth. Survival analysis is then performed on this conservative linked episode dataset, with censoring and sensitivity analysis used to account for unobserved or missed renewals.

Chronological Session Log

1. Built raw BOAMP acquisition logic and kept 2026 separate from the 2015–2025 historical corpus.
2. Created a full raw CSV and then a Grand Ouest raw CSV.
3. Created annual Grand Ouest Parquet files.
4. Built the EDA notebook and added a Grand Ouest section.
5. Read the internship guide and reframed the project around renewal detection and survival analysis.
6. Discussed why BOAMP has no ground truth and why manual benchmark evaluation is needed.
7. Created preprocessing and data-engineering notebooks.
8. Prepared the manual reference benchmark from three CSV files.
9. Generated renewal candidate pairs for benchmark anchors.
10. Compared baseline linkage algorithms.
11. Diagnosed baseline false positives and false negatives.
12. Implemented a stricter high-precision linkage iteration.
13. Added parameter-selection diagnostic plots.
14. Clarified that the project is currently at the linkage-method selection stage, not preprocessing or survival analysis.

15. Reframed the next optimal pipeline around episode construction, conservative linkage, and survival analysis.

Final Recommendation

The next implementation should be:

> Build procurement episodes from the engineered Grand Ouest digital notices, apply the frozen high-precision linkage rule to episode-level candidates, create a conservative renewal-link table, and then build the survival dataset.

The current linkage method is useful, but it should be described honestly as a conservative high-confidence renewal identification method, not a complete renewal detector.